

Section 6

INTRO TO SE

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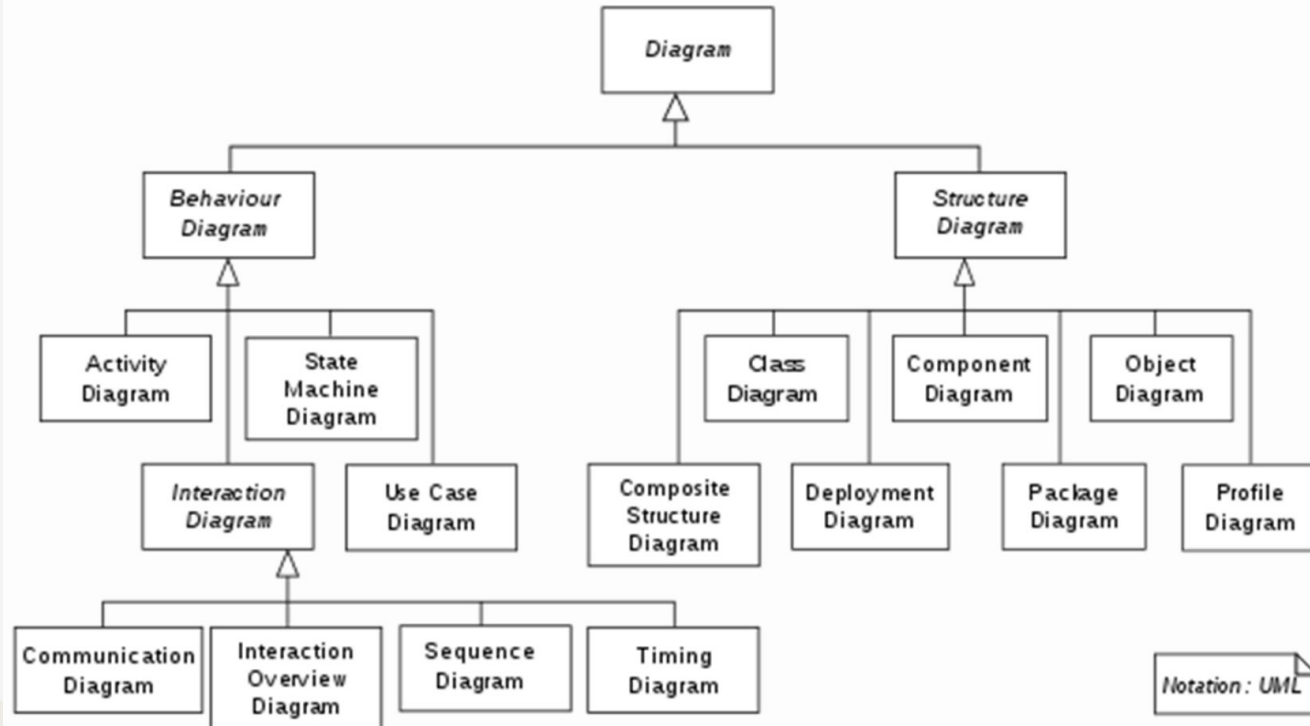
TA: Eng Salma Harb

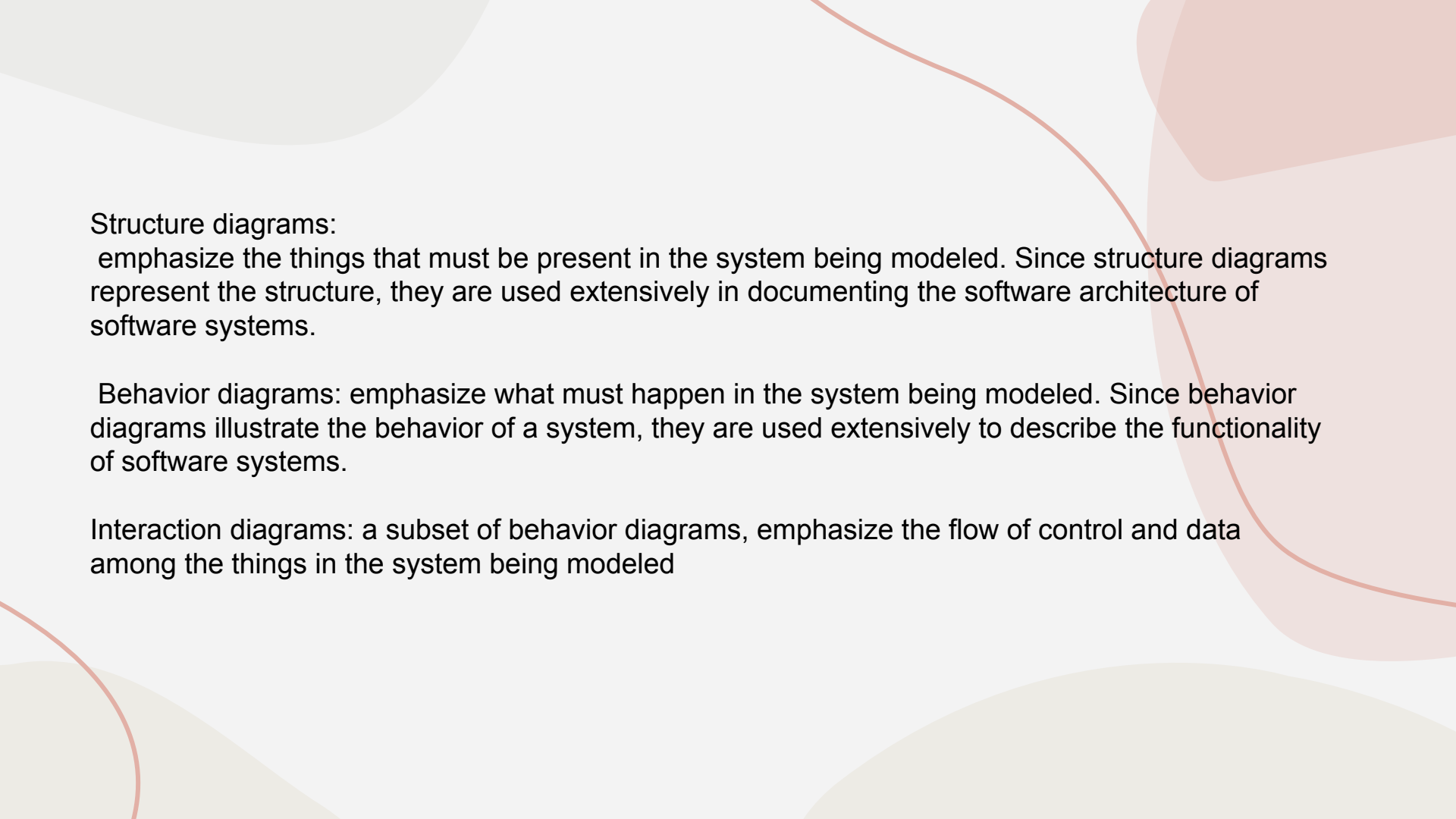
UML diagrams

UML stands for Unified Modeling Language. It's an international industry standard graphical notation used for describing, visualizing, constructing and documenting the artifacts of a software system.

UML 2 has many types of diagrams, which are divided into two categories. Some types represent structural information, and the rest represent general types of behavior, including a few that represent different aspects of interactions.

Classification of UML Diagrams





Structure diagrams:

emphasize the things that must be present in the system being modeled. Since structure diagrams represent the structure, they are used extensively in documenting the software architecture of software systems.

Behavior diagrams: emphasize what must happen in the system being modeled. Since behavior diagrams illustrate the behavior of a system, they are used extensively to describe the functionality of software systems.

Interaction diagrams: a subset of behavior diagrams, emphasize the flow of control and data among the things in the system being modeled

USE CASE DIAGRAM

A **Use Case Diagram** is a UML diagram that shows:

- **Actors** (users or external systems)
- **Use Cases** (actions or services the system provides)
- **Relationships** between them

Main Components:

1 Actor

- A person or system that interacts with the system.
- Drawn as a **stick figure**.

Example:

Web Customer, Payment Service ,Authentication Provider

It helps us understand **how users interact with the system**.

2 Use Case

- A function or service the system provides.
Drawn as an **oval**.
Example: View Items, Make Purchase, Client Register, Checkout

3 System Boundary

- The system being modeled.
- Drawn as a **rectangle** around use cases.

Example: **Online Shopping System**

4 Relationships

Association

- Line between actor and use case **<<include>>**
- One use case always includes another.

Example: Make Purchase → includes → Checkout

Online shopping

Web Customer actor uses some website to make purchases online.

Top level use cases are View Items, Make Purchase and Client Register.

View Items use case could be used by customer as top level use case if customer only wants to find and see some products.

This use case could also be used as a part of Make Purchase use case.

Client Register use case allows customer to register on the web site, for example to get some coupons or be invited to private sales.

Note, that Checkout use case is included use case not available by itself - checkout is part of making purchase.

Except for the Web Customer actor there are several other actors which will be described below with detailed use cases.

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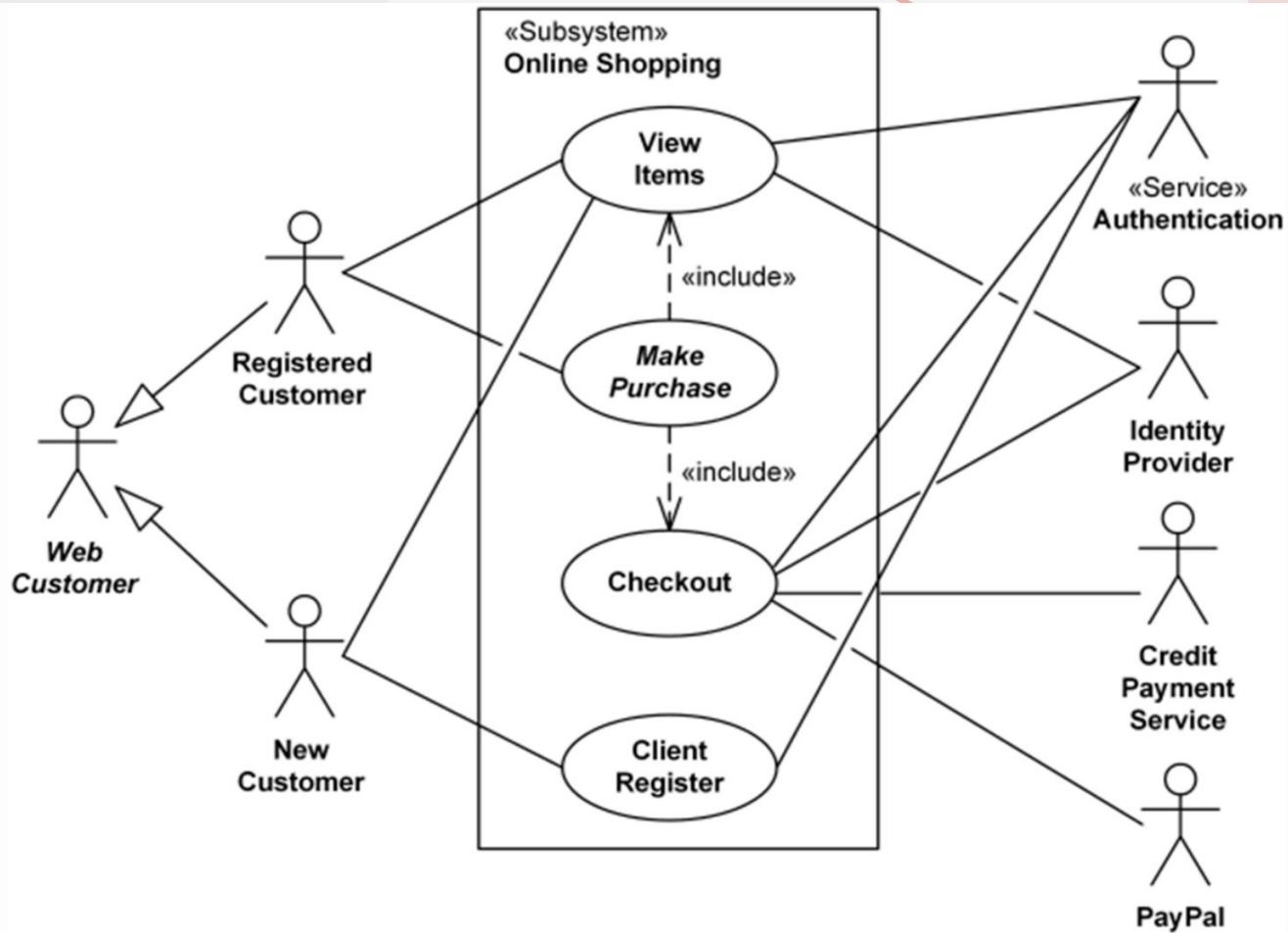
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Online shopping

- View Items use case is extended by several optional use cases - customer may search for items, browse catalog, view items recommended for him/her, add items to shopping cart or wish list.
- All these use cases are extending use cases because they provide some optional functions allowing customer to find item.
- Customer Authentication use case is included in View Recommended Items and Add to Wish List because both require the customer to be authenticated.
- At the same time, item could be added to the shopping cart without user authentication.



Activity diagram

An activity diagram visually presents a series of actions or flow of control in a system similar to a flowchart or a data flow diagram.

Activity diagrams are often used in business process modeling.

They can also describe the steps in a use case diagram.

Activities modeled can be sequential and concurrent.

In both cases an activity diagram will have a beginning (an initial state) and an end (a final state).

Activity Diagrams describe how activities are coordinated to provide a service which can be at different levels of abstraction.

Activity Diagram

Notations & Symbols

Initial State or Start Point A small filled circle followed by an arrow represents the initial action state or the start point for any activity diagram.



For activity diagram make sure the start point is placed in the top left corner of the first column.

Activity Diagram

Notations & Symbols

Activity or Action State

An action state represents the non-interruptible action of objects.

You can draw an action state using a rectangle with rounded corners.



Activity Diagram

Notations & Symbols

Action Flow

Action flows, also called edges and paths, illustrate the transitions from one action state to another.

They are usually drawn with an arrowed line.



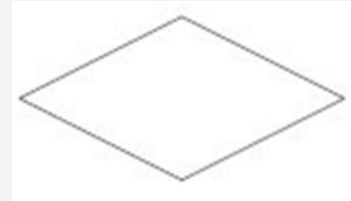
Activity Diagram Notations & Symbols

Decisions and Branching

A diamond represents a decision with alternate paths.

When an activity requires a decision prior to moving on to the next activity, add a diamond between the two activities.

The outgoing alternates should be labeled with a condition or guard expression. You can also label one of the paths "e

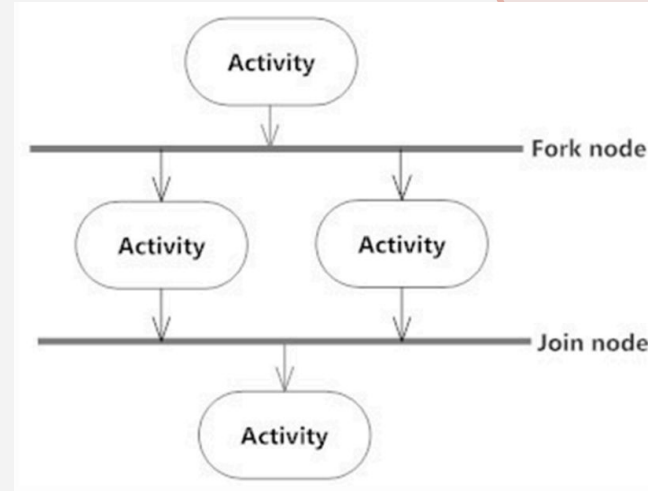


Activity Diagram Notations & Symbols

Synchronization

Fork Node: Split behavior into a set of parallel or concurrent flows of activities (or actions)

Join Node: Bring back together a set of parallel or concurrent flows of activities (or actions)



Activity Diagram

Notations & Symbols

Final State or End Point

An arrow pointing to a filled circle nested inside another circle represents the final action state.



